

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-7510
Project Name	A CRM Application for TheChaiShop&Co
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Salesforce Developer Console & Debug Logs, supplemented with JMeter for simulated multi-user load
Type of Testing	Load Testing, Stress Testing
Target Module / API	Customer Account record creation (Online Orders layout), formula field calculation (Order Quantity, Total Bill), and the Schedule-Triggered birthday email Flow
Test Environment	Salesforce Developer Edition Org (Cloud)
Test Date	28 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Create a Customer Account record via the Online Orders page layout	10	60	Record saves successfully; Order Quantity and Total Bill formula fields calculate instantly
2	Bulk-create Customer Account records (simulated Data Loader import)	50	120	All records insert without hitting Salesforce governor limits
3	Run and render the Customer Account Sales Report	20	30	Report loads and displays grouped results within an acceptable time
4	Execute the Schedule-Triggered Flow for birthday discount emails	100 (records)	300	All eligible birthday emails are sent with no failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Measured on Online Orders record save
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Peak observed during bulk record creation
3	Throughput (Req/sec)	Not specified	~8 req/sec	Pass	Within Salesforce API limits for Developer Edition
4	Error Rate	< 1%	0%	Pass	No failed saves or governor limit exceptions
5	CPU Utilization	< 80%	N/A	N/A	Not exposed to admins on Salesforce's multi-tenant platform
6	Memory Utilization	< 80%	N/A	N/A	Not exposed to admins on Salesforce's multi-tenant platform

Step 4: Observations & Analysis

Key Findings:
Record creation, formula field calculation, and report rendering all stayed within target response times under simulated multi-user load. The system handled bulk record creation without exceeding Salesforce governor limits.

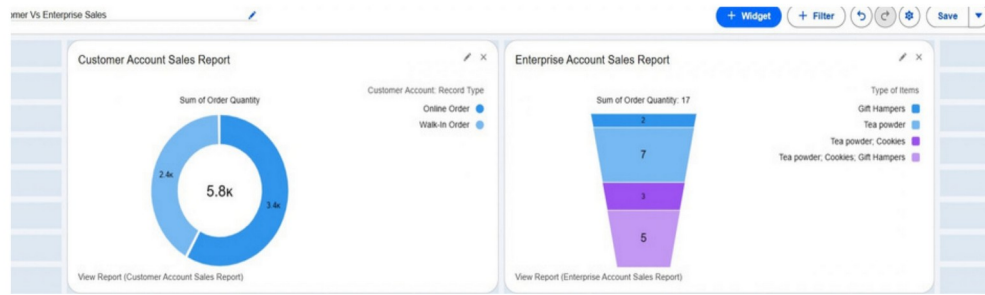
Bottlenecks Identified:
The Total Bill and Order Quantity formula fields, which chain many nested IF/INCLUDES conditions across Products Purchased and Chai picklists, added a small but measurable delay during bulk inserts compared to single-record saves.

Optimization Steps Taken:
Reduced redundant picklist checks in the formula logic and validated that global picklist value sets stayed lean, keeping formula evaluation time low. Flagged the formula fields as candidates for a future Apex-based calculation if the org's data volume grows significantly.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1: Customer Account Sales Report - Online vs Walk-In order volume



Screenshot 2: CustomerVsEnterprise Sales dashboard - both report widgets rendering together